

Nicholas Chin Bo You

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Education

Foon Yew High School	<i>Jan 2018 - Nov 2023</i>
University of Utah	<i>August 2024 – Present</i>
<i>BS in Physics with Computational Physics and Astronomy Emphasis</i>	
<i>BS in Applied Mathematics</i>	
◦ CGPA: 3.896	
◦ Dean’s List Fall 2024, Spring 2025, Fall 2025	

Selected Research & Publication

Searching for Photospheric Pulsation Modes of β Canis Majoris with Stellar Intensity Interferometry
Refining SII data-analysis workflows for the β Cephei variable β Canis Majoris to extract modulation signatures associated with photospheric pulsations.
Efficient Quantum State Preparation for Image Encoding
Studying sparsity-aware state-preparation methods for image encoding to reduce circuit complexity by exploiting the nonzero structure of sparse images.
Shared Funding Constraints and Variational Equilibrium in Option Dealer Hedging
Formulated option-dealer hedging with shared funding limits as a generalized Nash equilibrium problem and analyzed its variational equilibrium characterization.
Measurement of the photosphere oblateness of γ Cassiopeiae via Stellar Intensity Interferometry with the VERITAS Observatory A. Archer et al 2025 ApJ 995 191 🔗
<i>A. Archer, J.P. Aufdenberg, ..., N. B. Y. Chin, ..., The VERITAS Collaboration</i>

Presentations

Quantum Computing Focused Reading Group	Spring 2026
◦ Affiliated with NSF Research Training Group	
APS Four Corners Section 2025 Annual Meeting	Fall 2025
University of Utah Department of Physics and Astronomy Annual Symposium	Spring 2025

Skills

◦ Programming Languages: Python, C/C++, Rust, Java, Bash	
◦ Web & Markup: LaTeX, HTML, CSS	
◦ Scientific Computing : CERN ROOT, Jupyter Notebook, NumPy, Matplotlib, scikit-learn, MATLAB	
◦ Quantum Computing: Qiskit	
◦ Operating Systems: Linux / Unix	

Relevant Coursework

Intro to Quantum Computers - Graduate Course	In Progress
Enhanced Intro to Differential Equations - Honors Course	In Progress
Modern Physics	In Progress
Survey of Numerical Analysis	In Progress
Enhanced Linear Algebra - Honors Course	B+
Foundations of Astronomy	A
Accelerated Object-Oriented Programming	A-
Physics I, II	A
Physics I Lab	A
Calculus I, II, III	A

Certificates and Achievements

Summer Undergraduate Research Fellowship (SURF)	Summer 2026
Thomas J. Parmley Scholarship Endowment Fund	Fall 2026 - Spring 2027
MIT iQuHACK 2026 - BlueQubit Problem	World Rank 35, 2nd Highest Score
Undergraduate Research Opportunity Program Award (UROP)	Spring 2026
Talent Scholarship	Fall 2025 - Spring 2026
Chen Jingrun's Cup Secondary School Mathematics Competition 2021	National Rank 81st
Chen Jingrun's Cup Secondary School Mathematics Competition 2023	High Distinction
OpenAI Codex Hackathon	1 st runner up
IELTS 2024	6.5
MATLAB Onramp	Course Completion Certificate

Experience

Research Assistant

Measuring Stellar Sizes using ACT arrays as Intensity Interferometers

Salt Lake City, UT

Nov 2024 – Present

- Working with Professor LeBohec and Professor Kieda on refining data analysis process using CERN ROOT and C++.
- Developed C++/ROOT library to streamline analysis processes.
- Implemented an FPGA-to-ROOT file conversion tool.
- Measuring photosphere modulations of variable stars.

Undergraduate Student Advisory Committee - Chair

U of U Physics Department

U of U

Oct 2025 – Present

- Expanded committee membership by 300% through active recruitment and outreach initiatives.
- Organized a department-wide social event in collaboration with the Society of Physics Students (SPS), attracting 30+ participants across undergraduate, graduate, and non-physics programs.
- Led a student town hall attended by 40+ students to collect and communicate departmental feedback to faculty leadership.

Blockchain Club - Officer

Fintech Club - Treasurer

U of U Finance Department

U of U

Jan 2025 – Jun 2025

Jul 2025 – Present

- Developed educational content introducing blockchain, frontier science, and emerging computational technologies.
- Authored educational articles for the club's official website.
- Contributed regularly to the *FinTech Friday* weekly newsletter.

UCubesat Club - CFO

U of U ECE Department

U of U

Jan 2025 – Present

- Telemetry Team; Ground Station Subsystem Team Lead.
- Working on building a ground station to contribute to the SatNOGS Network.

Engineering Possibilities Summit Scholar

Goldman Sachs

Jan 2025 - Dec 2025

Apprentice Management

Rhapsody Youth Ensemble

Johor Bahru, Malaysia

2 projects (Approximately 6 months)

- Established the foundational framework for ensemble's operations and concert planning.

Co-Chairman and President of Education

Foon Yew Symphony Orchestra

Johor Bahru, Malaysia

June 2022 - May 2023

- Utilized recording and mixing skills to support the orchestra during the COVID period, facilitating virtual performances
- Organized and conducted more than 10 hours of online courses on different subjects weekly to maintain member engagement and skill development.
- Designed programs to help students foster interest and performing opportunities. Approximately 80% of members increased 40% more performing opportunities annually

Assistant Luthier and Sales Assistant

Melodia String

Johor Bahru, Malaysia

Jan 2024 - Aug 2024

- Performed walk-in instrument maintenance and basic repairs, including string replacement, setup adjustments, and minor restorations
- Studied and completed projects up to an intermediate level luthiery techniques
- Maintained workshop equipments and tools

Recording Studio Assistant

Goldstone Music

Johor Bahru, Malaysia

Jan 2024 - Aug 2024

- Assisted with recording session setup, including microphone placement, signal routing, and basic audio equipment configuration
- Provided technical and logistical support during live stage performances and events
- Studied music arrangement and audio recording techniques through hands-on observation and guided practice

Projects

Option Pricing using Quantum Computers

[*Option-Pricing-on-Quantum-Computer*](#) 

- A from-scratch implementation of Stamatopoulos et al. (2020). "Option Pricing using Quantum Computers"

A Guide to Quantum Computing

[*A-Guide-to-Quantum-Computing*](#) 

- A collection of notes and implementations aimed at building intuition for quantum computing concepts without requiring prior knowledge of quantum mechanics.

A Complete DNS Server

[*DNS-Server-in-RUST*](#) 

- Developed a fully functioning DNS Server that handles DNS queries, resolves domain names, and returns IP addresses following standard DNS protocol specifications with Rust.